

Proposal

Advisory services Substation EMC

Management

SC&M II V.O.F.

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About this document

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for DNV Netherlands B.V.

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1 INTRODUCTION

1.1 Project Context & Background

This document presents DNV's proposal to provide specialist Electromagnetic Compatibility (EMC) advisory services to SC&M II V.O.F. This is in direct response and in compliance with to your "Inkoopspecificatie EMC Advies" (Purchase Specification for EMC Advice), reference 2508-0182, for the Simonshaven 380kV substation expansion project.

DNV understands the critical nature of the planned expansion (sub-projects DP2-4). The project's primary drivers are twofold: firstly, to enable the connection of the new Amaliahaven substation, which is essential for the 2GW IJmuiden Ver offshore wind connections; and secondly, to facilitate a new 150kV supply for the vital Botlek and Pernis industrial areas.

The core technical challenge is the successful integration of significant new high-voltage assets—including new transformers, compensation coils, and the infrastructure for bays 4 and 5—into a live and operational high-voltage substation. Ensuring this expansion is executed without compromising the operational integrity of the existing station (DP1) or surrounding infrastructure is paramount. This requires a robust, verifiable, and holistic EMC compliance strategy that satisfies the stringent requirements of the Transmission System Operator, TenneT.

1.2 DNV's Role as Your Independent EMC Partner

As an independent expert and principal consultant, DNV's role is to act as your trusted partner, providing the specialist knowledge required to navigate the complexities of EMC compliance for this project. Our objective is to work collaboratively with the SC&M team to de-risk the design and implementation phases, ensuring a technically sound and fully compliant outcome.

We base our analysis and recommendations firmly on industry standards and best practices, as defined by TenneT's own standards (including the EMC eisen-zone concept) and relevant international norms (NEN, IEC).

Our ultimate goal is to provide you with a clear, actionable, and auditable EMC compliance framework that not only ensures the safe and reliable operation of the expanded Simonshaven substation but also facilitates a smooth and efficient acceptance process by TenneT.

2 REQUIREMENTS

2.1 Alignment with "Inkoopspecificatie EMC Advies" (ref: 2508-0182)

DNV has performed a detailed analysis of the scope of work (*Ontwerpscope*) as defined in Section 2.1 of your purchase specification. We understand that the core of your request is to procure independent, expert-level EMC services to ensure the successful integration of the DP2-4 expansion at the Simonshaven 380kV substation.

We confirm our understanding that the required scope of work encompasses the following key tasks:

1. **Baseline Assessment:** To conduct a site survey (*schouw/nulinspectie*) of the existing station, document the findings, and perform a critical review of the existing "Beoordeling EMC studie SMH380" (doc. 01350-01-00075).
2. **EMC Management Plan:** To develop a comprehensive EMC Management Plan (*EMC Beheersplan*) for the DP2-4 expansion, detailing the compliance strategy, EMC zoning, a risk-based EMC matrix, and specific mitigation measures, following the structure of the existing DP1 plan (doc. 01350-01-00074).
3. **Integration Measures:** To define and provide justification for the EMC measures necessary to ensure the seamless and correct functioning of the existing DP1 installation together with the new DP2-4 assets.

4. **Application of TenneT Standards:** To specifically interpret and apply TenneT's generic "E04 Uitwerking EMC eisen-zone concept" document to the Simonshaven site, clearly identifying applicable requirements and justifying any necessary deviations within the EMC Management Plan.
5. **Field Measurement:** To determine the baseline electric and magnetic field levels (*Vaststellen E-/M-veld*) at the station's perimeter fence.
6. **Verification of Studies:** To review and verify the provided influence studies (docs. 1287695 E01, E02, E03) against the specific requirements outlined in VS2, and to formally document the findings in a "Rapport Controle Beïnvloedingsstudie".

We further understand that these activities must culminate in the delivery of four specific, formal reports as the primary project deliverables.

2.2 Key Project Objectives & Challenges

Beyond the specific tasks, we recognize that the ultimate goal is to achieve **demonstrable EMC compliance** for the entire, expanded Simonshaven substation. The primary objectives are:

- **Technical Integrity:** To ensure the safe, reliable, and interference-free operation of all primary and secondary systems within the substation.
- **Regulatory & Client Acceptance:** To achieve acceptance by TenneT is part of DNV's scope of work. DNV will produce a robust and auditable verification file that provides TenneT with the necessary assurance to formally accept the completed works.

The primary technical challenge lies in bridging the gap between the existing, operational DP1 installation and the new DP2-4 design. This requires a detailed analysis of the electromagnetic interactions between these phases and the definition of practical, cost-effective engineering solutions (for earthing, shielding, cabling, etc.) that guarantee compatibility. Our approach is designed to meet this challenge head-on.

3 PROPOSED SCOPE OF WORK & METHODOLOGY

3.1 Overall Phased Approach

DNV will execute this project in a structured, multi-phase approach designed for clarity, efficiency, and technical rigor. Our methodology is built on four key pillars:

1. **Baseline Definition:** Understanding the existing "as-is" situation through on-site inspection and documentation review.
2. **Analysis & Verification:** Performing the core engineering analysis to identify risks and verify compliance against all applicable standards.
3. **Solution Development:** Defining concrete, practical mitigation measures within a comprehensive management plan.
4. **Formal Reporting:** Consolidating all findings and evidence into the formal deliverables required for project verification and acceptance.

The following tasks detail our execution plan for each phase.

3.2 Task 1: Baseline Assessment & Site Survey

(Addresses Inkoopspecificatie Scope Items 1 & 5)

The first step is to establish a definitive technical baseline.

- **3.2.1 On-Site Inspection (Schouw/Nulinspectie):** A senior DNV EMC specialist will conduct a thorough on-site survey of the existing Simonshaven substation (DP1). This technical inspection will focus on key EMC aspects, including the condition and implementation of the earthing grid, bonding practices, cable routing and segregation, installation practice employed and the physical state of equipment. The findings will be documented with photographic evidence and compared against the existing design documentation and the conclusions of the previous EMC study (doc. 01350-01-00075). Note: DNV assumes when excavation is required for inspection, this will be arranged by the Client.

3.3 Task 2: Review and Verification of Existing Studies

(Addresses Inkoopspecificatie Scope Item 6)

We will conduct a formal review of the client-provided influence studies to verify their conclusions and applicability to the expanded project scope.

- **3.3.1 Analysis of TenneT and NUON Influence Studies:** We will perform a critical review of the "Beïnvloedingsstudie omgeving SMH380" and the "Vergelijking magneetvelden Simonshaven" reports. Our analysis will validate the assumptions, methodologies, and conclusions regarding inductive interference, Ground Potential Rise (GPR), and compliance with both the 100 μ T public exposure limit and the 0.4 μ T precautionary policy zone for sensitive areas.
- **3.3.2 Verification Against VS2 Requirements:** The findings and conclusions of the studies will be systematically checked against the specific contractual requirements: VS2-req-0652, VS2-req-0653, VS2-req-0654, and VS2-req-0659. The results of this verification will be documented in the "Rapport Controle Beïnvloedingsstudie".

3.4 Task 3: Development of the EMC Management Plan (EMC Beheersplan)

(Addresses Inkoopspecificatie Scope Items 2, 3 & 4)

This is the primary engineering task, resulting in the core guidance document for the project.

- **3.4.1 Impact Analysis of DP2-4 Expansion:** We will analyze the new single-line diagrams, layout drawings, and equipment specifications for the DP2-4 expansion. This analysis will identify new potential sources of electromagnetic disturbance, new sensitive equipment (victims), and the coupling paths between them, assessing the overall impact on the station's EMC environment.
- **3.4.2 Application of TenneT's EMC Zone Concept:** Using TenneT's standard "Uitwerking EMC eisen-zone concept" (doc. 1287698 E04) as our guiding document, we will develop a site-specific EMC zone plan for the fully expanded Simonshaven station. This will define the boundaries between different EMC zones (e.g. EM zones aligned with lightning protection zones, LPZ 0, LPZ 1) and specify the interface requirements at each boundary. Any necessary deviations from the generic standard will be identified and technically justified.
- **3.4.3 Development of Mitigation Measures & EMC Matrix:** Based on the impact analysis and zoning plan, we will define a set of concrete, practical, and cost-effective EMC mitigation measures. These will include, but are not limited to:
 - Earthing and bonding strategy for new equipment and structures.
 - Requirements for cable routing, segregation, and shielding.

- Specifications for cable entry into the CDG and new shelters (EMC glands, frames).
- Lightning protection integration. Where the emphasis is on integration by means of references to standards and excludes the design of the external and internal lightning protection systems.

These measures will be summarized in an updated **EMC Matrix**, which provides a clear risk-based overview of potential interference paths and the corresponding protective measures.

3.5 Task 4: Final Reporting & Verification

(Addresses all resultaatdocumenten)

The final phase consolidates all work into the required formal documentation.

- **3.5.1 Authoring of Key Deliverables:** All findings from the preceding tasks will be professionally authored and compiled into the four distinct reports specified in your request.
- **3.5.2 Verification Traceability Matrix:** A key component of the *Verificatierapportage* will be a traceability matrix. This matrix will provide a clear and auditable trail, cross-referencing every applicable EMC requirement from the VS1 and VS2 requirement sets to the specific section, drawing, or mitigation measure in our EMC Management Plan that demonstrates compliance. This is a critical tool for ensuring a smooth acceptance process with TenneT.

4 DELIVERABLES

Upon successful completion of the scope of work detailed in Section 3.0, DNV will deliver a comprehensive set of formal reports. These documents are designed to provide a complete and auditable EMC compliance file for the Simonshaven DP2-4 expansion project, fulfilling all documentation requirements as stipulated in Section 2.1 of the "Inkoopspecificatie EMC Advies".

The following four (4) key documents will be delivered in Dutch, in digital (PDF) format:

1. **Rapport Schouw/nulinspectie (Survey/Baseline Inspection Report)**
This report will present the findings of the on-site survey and the baseline E/M field measurements. It will include photographic documentation, a summary of the "as-is" EMC conditions of the existing station, and a formal assessment of the previously conducted EMC study for DP1.
2. **DP2-4 EMC Beheersplan (DP2-4 EMC Management Plan)**
This will be the primary guidance document for ensuring EMC compliance throughout the design and execution phases. It will contain the site-specific EMC zoning plan, the updated EMC risk matrix, and a detailed description of all required mitigation measures for earthing, bonding, shielding, and cabling for the new DP2-4 assets and their interfaces with the existing station. This will be performed, keeping in mind that the EMC management of DP1 has to be considered as well. For example, proposals will be made to improve the EMC on the DP1 assets as well.
3. **Verificatierapportage (Verification Report)**
This report will provide the formal demonstration of compliance. It will contain the verification traceability matrix, which systematically proves that all applicable EMC requirements from the VS1 and VS2 sets have been met by the proposed design and the measures outlined in the EMC Management Plan.
4. **Rapport Controle Beïnvloedingsstudie (Report on the Control of the Influence Study)**
This report will document the results of our verification of the existing influence studies (TenneT and NUON reports) against the specific contractual requirements (VS2-req-0652, VS2-req-0653, VS2-req-0654, and VS2-req-0659).

5 SCHEDULE

5.1 Proposed Project Milestones

The project will be structured around the following key milestones, starting from the official project kick-off date (T0):

Milestone	Activity	Proposed Timing
M1	Project Kick-off Meeting & Final Information Request	T0 + 1 week
M2	On-Site Survey at SMH380	T0 + 3 weeks
M3	Submission of Draft Reports for SC&M Review	T0 + 10 weeks
M4	Receipt of Consolidated Feedback from SC&M	T0 + 12 weeks
M5	Final Delivery of All Formal Reports	T0 + 14 weeks

This schedule allocates approximately two months for the core engineering and analysis work, followed by a collaborative review cycle to ensure the final deliverables fully meet your expectations and are ready for submission to TenneT. DNV will work towards fast tracking the work to deliver a first draft in week 2 of December 2025. However, this is dependent on timely sharing of information and contract agreement dates.

5.2 Dependencies and Assumptions

The successful execution of this schedule is based on the following key dependencies and assumptions:

- **Timely Information Provision:** All required project documentation, including the complete set of design documents for DP2-4 (drawings, equipment lists, etc.) and any other documents listed in the *Inkoopspecificatie*, will be made available to DNV at the project kick-off (M1).
- **Site Access:** DNV personnel will be granted timely and safe access to the Simonshaven 380kV substation to perform the on-site survey and measurements as scheduled (M2). All necessary site inductions and permits will be arranged by SC&M.
- **Efficient Communication:** A single point of contact will be designated within the SC&M project team to streamline communication and facilitate prompt responses to technical queries.
- **Review Period:** The schedule assumes a review and feedback period of one (1) week for SC&M (between M3 and M4).
- **Meetings:** DNV will support with bi-weekly meetings to provide feedback on progress.

DNV is committed to proactively managing the schedule and communicating any potential risks or delays promptly. By working in close partnership, we are confident that we can meet the 31 October 2025 deadline.

5.3 Project team

DNV will assign a dedicated team of experts with extensive experience in high-voltage substation EMC and specific knowledge of TenneT's standards.

- **Project Manager & Lead EMC Consultant (Peet Schutte and Pieter Pretorius):** Peet will be your single point of contact, providing overall project oversight and senior technical leadership. He will work with Pieter Pretorius who will be responsible for the strategic direction of the EMC plan and the final review of all deliverables. An alternative PM might be assigned at project kickoff, which is determined by resource availability at the time.

- **Supporting EMC Engineers:** A team of DNV consultants will support the project by performing detailed document reviews, developing the EMC matrix, and conducting the E/M field modeling.

DNV maintains a large, global pool of highly qualified engineers. Should the project schedule or scope change, we can mobilize additional experienced resources to ensure all commitments are met without compromising quality or timelines.

6 REMUNERATION, TERMS OF INVOICING, COSTS/EXPENSES

The project will be delivered on a **lump sum basis** for the complete scope of work as described in this proposal.

Total Lump Sum Price: EUR 49 830,00.

Rates and prices provided in this proposal are quoted in Euro and are exclusive of VAT or any other similar taxes.

6.1 Invoicing and Payment

DNV proposes a milestone-based invoicing schedule that aligns payments with the successful completion and submission of key project deliverables. This ensures a transparent and fair commercial process for both parties.

Invoices will be submitted as follows, based on a percentage of the total contract value:

Payment Stage	Milestone / Deliverable	% of Total Contract Value
1. Mobilisation	Upon receipt of your formal Purchase Order (PO).	10%
2. First Progress Payment	Upon submission of the draft <i>Rapport Schouw/nulinspectie</i> for your review.	30%
3. Second Progress Payment	Upon submission of the draft versions of the following three reports for your review: • <i>DP2-4 EMC Beheersplan</i> • <i>Verificatierapportage</i> • <i>Rapport Controle Beïnvloedingsstudie</i>	40%
4. Final Payment	Upon final delivery of all four (4) approved project deliverables, incorporating your feedback.	20%
Total		100%

All invoices are payable within 30 days of the invoice date.

Prices are exclusive of any form of withholding tax, sales tax, value added tax (VAT), goods & services tax, custom, excise & stamp duties and/or any other similar taxes including any surcharges levied thereon (hereinafter: Taxes). At contract negotiations it will be determined whether any Taxes will be applicable under the mandatory laws.

The contact person(s) shown in the table below will be pleased to answer any queries you may have in respect of this proposal.

Table 4-1 Contact information

Name	Title	Telephone No.	Email
Peet Schutte	Consultant	+31 6 4219 5249	Peet.schutte@dnv.com



7 CONTRACTUAL

DNV will perform under the General Purchase Conditions TBI 2022 under the following conditions:

Notwithstanding anything in the Contract or otherwise to the contrary, to the extent permitted by law, DNV's maximum cumulative liability arising out of or related to the Contract, whether arising in contract, in tort (including negligence), by indemnity or otherwise, shall be limited to the lesser of 10 times the contract value or USD 300.000.

8 ACCEPTANCE

We believe this is an appropriate proposal and we are looking forward to receiving your order. In case you have any questions please contact Peet Schutte, email: peet.schutte@dnv.com tel. 0642195249

Please send Section I – Cover Letter of the SFA duly countersigned for acceptance to Salesdesk@dnv.com or by mail to:

DNV Netherlands B.V.
Attn. Department Support (B50)
P.O. Box 9035
6800 ET ARNHEM
The Netherlands



About DNV

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